Trigonometric Approach

3R. Planner Robot

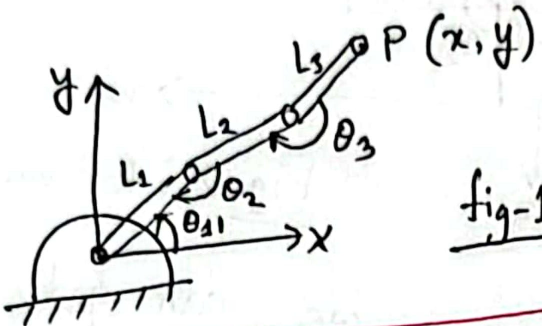
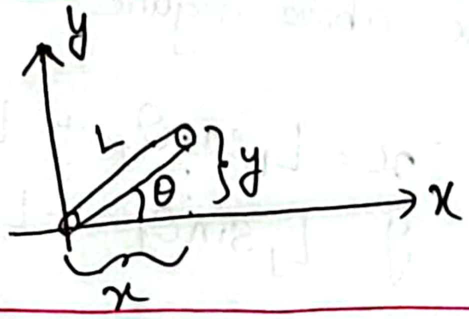


fig-1: 3R Planner Robot

We know, for any link,



So, $x = L\cos\theta$
$y = L\sin\theta$

Note 1

But, the robot don't have angles with x-axis or Y-axis (fig-1: 3D Planner). The angles are given with respect to the previous links.

Therefore, from θ_2 and θ_3 we will calculate the angles with respect to the x-axis by drawing triangles.

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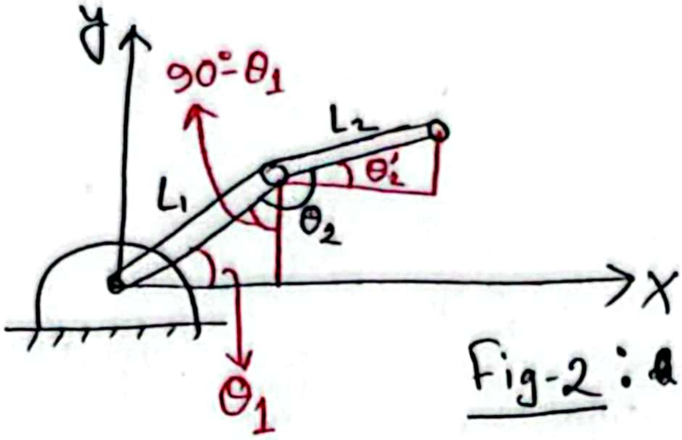


Fig-2 : Planner Robot

From the above figure, we can get,

$$x = L_1 \cos\theta_1 + L_2 \cos\theta_2' \quad \dots \quad \dots (1)$$
$$y = L_1 \sin\theta_1 + L_2 \sin\theta_2' \quad \dots \quad \dots (2)$$

$$\boxed{\begin{aligned} \theta_2' &= \theta_2 - 90° - (90° - \theta_1) \text{[From the figure]} \\ &= \theta_2 - 90° - 90° + \theta_1 \\ &= -180° + \theta_1 + \theta_2 \end{aligned}}$$

The resultant,

$$X = L_2 \cos(-180° + \theta_1 + \theta_2) + L_1 \cos\theta_1$$
$$Y = L_2 \sin(-180° + \theta_1 + \theta_2) + L_1 \sin\theta_1$$

Now, If the $\boxed{\theta_3 \text{ joint}}$ also added, you can get

$$X = L_1 \cos\theta_1 + L_2 \cos(-180° + \theta_1 + \theta_2) + L_2 \cos(-360° + \theta_1 + \theta_2 + \theta_3)$$
$$Y = L_1 \sin\theta_1 + L_2 \sin(-180° + \theta_1 + \theta_2) + L_2 \sin(-360° + \theta_2 + \theta_3)$$

[Verify by yourself]

An Extension:

Let say the angles are given from sides:

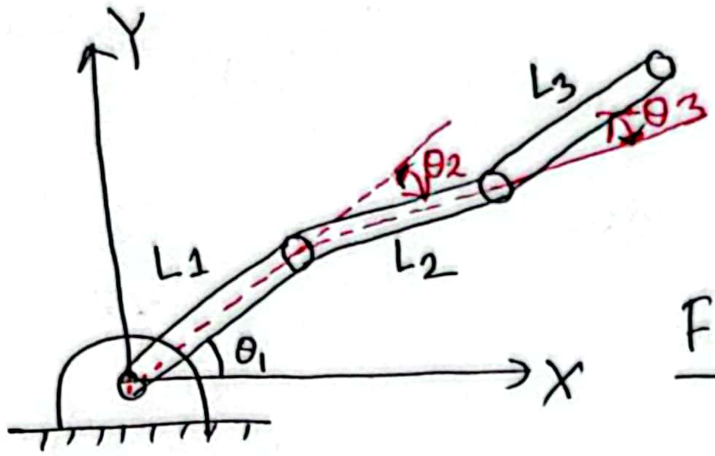


Fig-3:

Here the angles are given with respect to previous Link's length across.

Now, if we compare with Fig-1 or Fig-2 those angles were different.

So, if we want to use the previous equation: we want to convert this to previous formate.

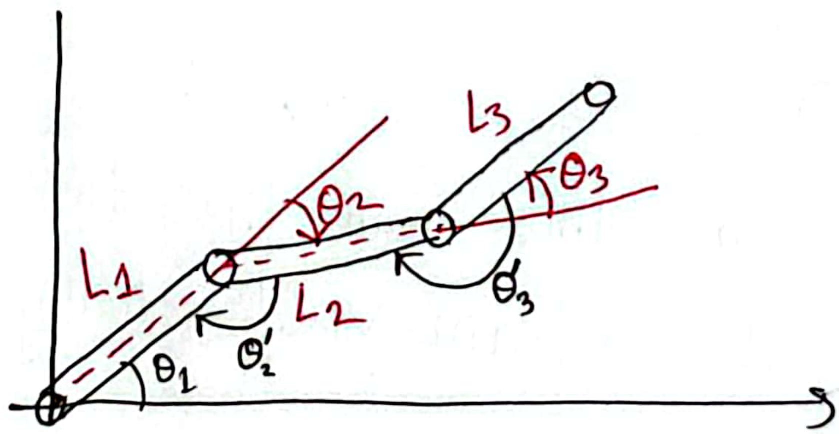


Fig-4

$$X = L_1 \cos\theta_1 + L_2\cos(\theta_2' + \theta_1 + 180^\circ) + L_3\cos(\theta_3' + \theta_2' + \theta_1 - 360^\circ) \quad (3)$$

Now,

$$\theta_2' = 180^\circ - \theta_2$$

$$\theta_3' = 180^\circ + \theta_3$$

So, if we plug this into the eq. (3),

$$X = L_1\cos\theta_1 + L_2\cos(\theta_1 - \theta_2) + L_3\cos(\theta_1 - \theta_2 + \theta_3)$$

$$Y = L_1\sin\theta_1 + L_2\sin(\theta_1 - \theta_2) + L_3\sin(\theta_1 - \theta_2 + \theta_3).$$

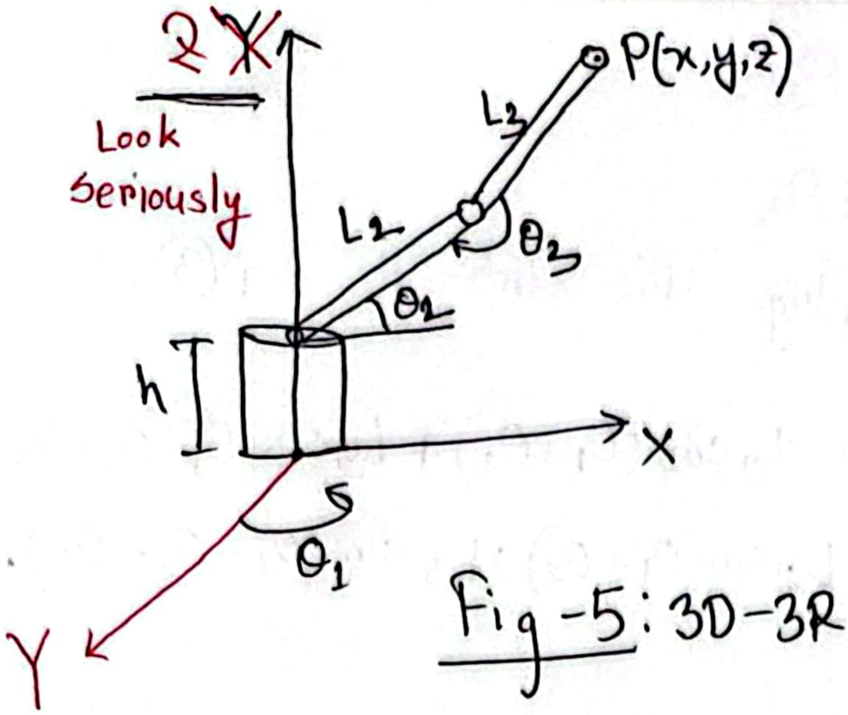
$\boxed{\text{Mind it}}$:

The θ_2 & θ_3 values are different from the previous math & the extension part.

Caveats:

3R Planner operate only in a single plane, do not operate in the 3 axis altogether.

3R - 3D Robot



This Robot's Base joint is connected differently than previous 3R planner Robot.

Fig-5: 3D-3R Robot

In this Robot, we can see link-2 and link-3 are 3R planner type.

As there are $\boxed{x, y, z}$, we need two a view-point for three axis values.

We will use two view-point.

1) Side-View point
2) Top-View point

SIDE VIEW:

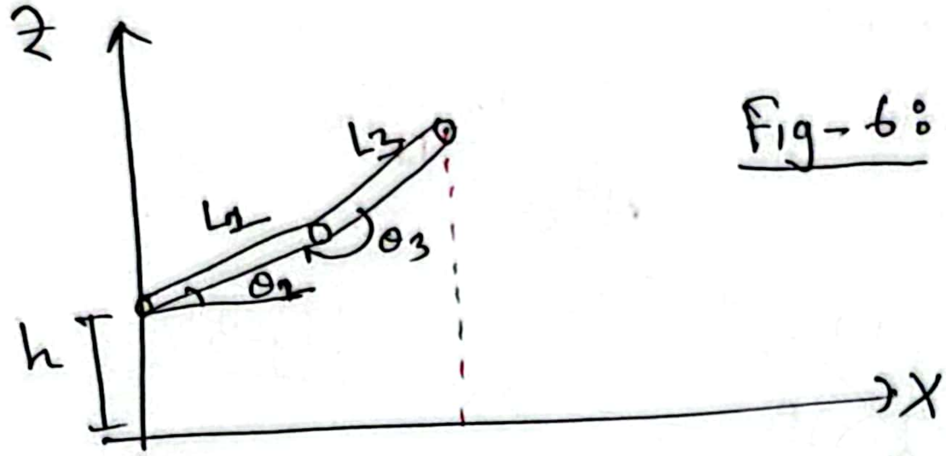
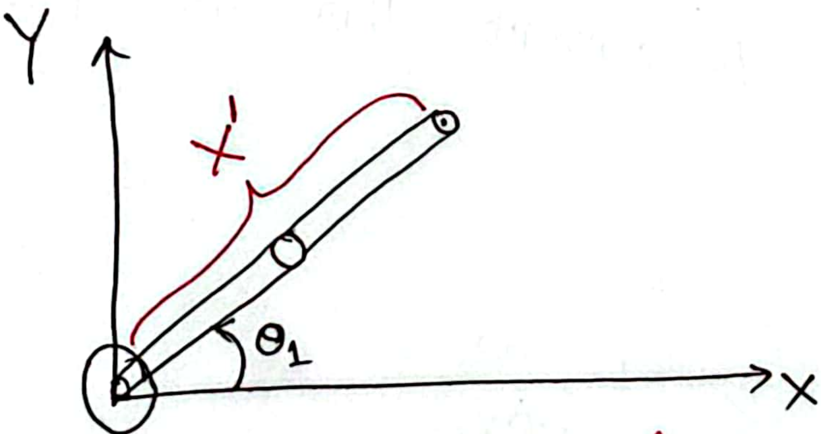


Fig-6: Side view

$$X = L_2 \cos\theta_2 + L_3 \cos(-180^\circ + \theta_2 + \theta_3)$$
$$z = L_2 \sin\theta_2 + L_3 \sin(-180^\circ + \theta_2 + \theta_3) + h$$

From Planner Robot Theory.

Top View:



When we view this from the top-view the value of X [SIDE View] actually not the x-value, rather the projection of arm values. So, the X value from side value is $X' = X$ [projection value]

So,

the real,

$$X = X'\cos\theta_1 = \left(L_2\cos\theta_2 + L_3\cos(-180^\circ + \theta_2 + \theta_3)\right)\cos\theta_1$$

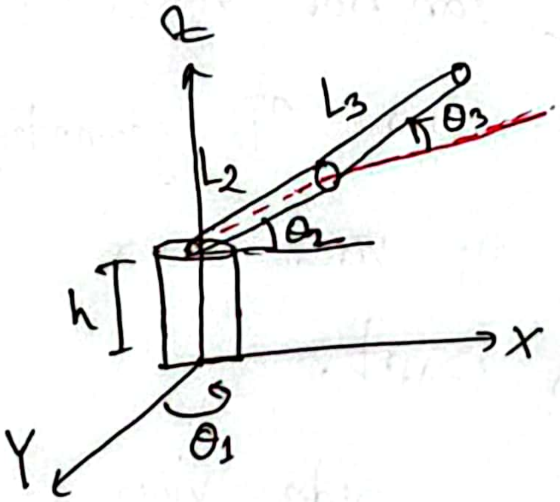
$$Y = X'\sin\theta_1 = \left(L_2\cos\theta_2 + L_3\cos(-180^\circ + \theta_2 + \theta_3)\right)\sin\theta_3$$

Finally,

$$Z = L_2\sin\theta_2 + L_3\cancel{\cos}\sin(-180^\circ + \theta_2 + \theta_3) + h$$

Extention

(1)

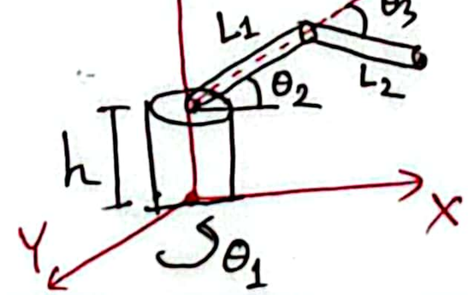


$$X = \left(L_2\cos\theta_2 + L_3\cos(\theta_2 + \theta_3)\right)\cos\theta_1$$

$$Y = \left(L_2\cos\theta_2 + L_3\cos(\theta_2 + \theta_3)\right)\sin\theta_1$$

$$Z = L_2\sin\theta_2 + L_3\sin(\theta_2 + \theta_3) + h$$

(2)



$X = ?$

$Y = ?$

$Z = ?$

Do it yourself.